

One Difference, Not Two Separate Intervals

Topic 6.8 · TODAY'S PROBLEM

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let $p_O - p_P$ mean Oak minus Pine.

Elena: “The two-sample z -interval is $(0.030, 0.270)$.”

Devon: “Subtract the farthest endpoints of separate 95% intervals. That gives $(-0.020, 0.320)$, so zero is plausible.”

Household samples

Town	Use	Other	n	N
Oak	77	63	140	2200
Pine	48	72	120	1800

FIRST TAKE (5 min, on your sheet)

Which method gives the requested 95% interval for $p_O - p_P$? Cite one design or numerical detail.

- DOK 1** (a) Define p_O and p_P and name the interval procedure for $p_O - p_P$.
- DOK 2** (b) Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.
- DOK 3** (c) Choose the warranted method. Use design, counts, variance structure, and zero's location. Explain why separate 95% intervals do not produce the requested coverage and how Devon's result misleads.

Video + follow-along: u6_lesson8_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

